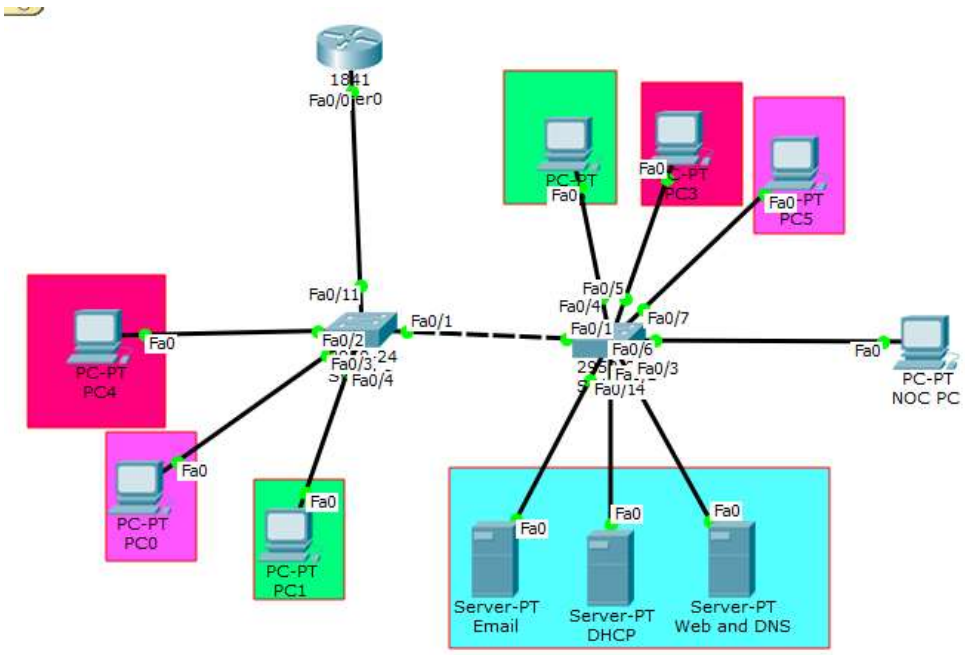
Packet Tracer Practical Assignment

Scenario Overview

TechNova Ltd. requires a secure enterprise network divided by departments, with a **centralized server VLAN** for core services: DHCP, Web/DNS, and Email.

As the network engineer, your task is to design and configure a full solution in **Cisco Packet Tracer** using:

- Department VLANs
- A **single dedicated Server VLAN** for all servers
- Inter-VLAN routing
- Static IPs for servers
- DHCP for client PCs
- DNS, Web, and Email service
- Switch security (port security, management access)
- Telnet-enabled remote switch access



Devices Required

- 1 Router
- 2 Switches
- 3 Servers: DHCP, DNS/Web, Email
- 6 PCs (2 per department)
- 1 Admin PC (for Telnet/management)

Configuration Tasks

1. VLAN Setup

VLAN Name	VLAN ID	Subnet
Admin Dept	10	192.168.10.0/24
HR Dept	20	192.168.20.0/24
IT Dept	30	192.168.30.0/24
Server VLAN	50	192.168.50.0/24
Management	99	192.168.99.0/24

- Assign PCs to their respective VLANs.
- Place **all servers in VLAN 50**.
- Use **trunk links** between switches and the router.

2. Inter-VLAN Routing

- Use **Router-on-a-Stick**: configure one subinterface per VLAN on the router.
- Ensure all departments can access the server VLAN.

3. Server IP Configuration (Manual)

- Place all 3 servers in **VLAN 50**
- Assign static IPs:

Server	IP Address
DHCP Server	192.168.50.10
DNS/Web Server	192.168.50.20
Email Server	192.168.50.30

4. DHCP Configuration

- DHCP server must provide IPs for:
  - VLAN 10 (Admin)
  - VLAN 20 (HR)
  - VLAN 30 (IT)
- Use correct gateway and DNS settings.
- **Do not use DHCP for servers or switch management.**

5. DNS + Web

- Configure `www.technova.local` to point to the Web Server.
- Enable HTTP.
- Test via browser from client PCs.

6. Email

- Enable SMTP and POP3.
- Create at least 2 users (e.g., admin@technova.local, hr@technova.local).
- Send and receive test emails.

7. Switch Security

- Set:
  - Hostname
  - MOTD banner
  - Console, VTY, and enable secret passwords
- Configure **VLAN 99** as the management VLAN.
- Assign IP address on VLAN 99.
- Enable **Telnet** access.

8. Port Security

- On all user access ports:
  - Allow only **1 sticky MAC**
  - Violation mode: **restrict**
  - Demonstrate enforcement by connecting different device

9. Telnet Access

• The NOC PC (in VLAN 99) must be able to Telnet into the switches using their management IPs.

• Use telnet 192.168.99.x from NOC PC.

Student Must Demonstrate

- PCs receive IP via DHCP
- DNS resolves correctly
- Website accessible by domain
- Email successfully sent
- Telnet connection to switches works
- Port security enforcement visible via `show` commands

Marking Scheme (100 Marks)

Section	Criteria	Marks
Topology Setup	All required devices, correct physical connections	10
VLAN Configuration	Department VLANs + dedicated **Server VLAN (VLAN 50)**	5
	Ports assigned to correct VLANs	5
	Trunking between switches/router	5
Router-on-a-Stick	Subinterfaces + correct encapsulation + IPs	10
Server IPs (Static)	All servers use correct IPs in VLAN 50	10
DHCP Server	Pools for 3 client VLANs, proper gateways	5
	Clients receive correct IPs	5
DNS/Web	DNS resolves, website loads via domain	10
Email	Users added, email sent/received	10
Switch Setup	Hostname, passwords, IP on VLAN 99	5
Telnet Access	Successful from admin/NOC PC	5
Port Security	Sticky MAC, restrict mode, tested	10